Section 15.1b: Integration Review

Multiple Integrals

MTH 310
Calculus III

Why Are We Reviewing Integration?

Every double and triple integral you'll compute this semester eventually reduces to **single-variable integrals**. If you're rusty on these, the new ideas get buried under algebra struggles.

Today's goal: **practice your integration toolkit** so the rest of Chapter 15 goes smoothly.

You have a **formula sheet** with all the identities and formulas you'll need. Today is about learning to *use* them:

- u-substitution — the workhorse of integration
- Trig identities — for powers of sin/cos
- Integration by parts — for products of different function types
- Trig substitution — for square roots that resist u-sub
- Recognizing impossible integrands

The formulas are on your sheet. The **skill** is knowing which one to reach for and how to apply it.

u-Substitution: The Most Important Technique

When to use it: You see a composition $f(g(x))$ and its derivative $g'(x)$ nearby.

The steps:

1. Choose $u =$ (the “inner function”)
2. Compute $du = g'(x)\,dx$
3. Rewrite **everything** in terms of u and du
4. Integrate, then substitute back

The key skill: **spotting the chain rule in reverse**. If you see $g'(x)$ multiplied by a function of $g(x)$, let $u = g(x)$.

Example 1: u-Substitution with Exponentials

Example 1

Evaluate $\displaystyle\int y\, e^{y^2}\, dy$.

This exact integral appears when evaluating $\iint_R e^{y^2}\, dA$ after reversing the order of integration (Section 15.2).

Let $u = y^2$. What is du? Rewrite and integrate.

Choose u: Let $u = y^2$, so $du = 2y\, dy$, which means $y\, dy = \frac{1}{2}\, du$.

Substitute:

$$\int y\, e^{y^2}\, dy = \int e^u \cdot \frac{1}{2}\, du = \frac{1}{2} e^u + C = \frac{1}{2} e^{y^2} + C$$

$\displaystyle\int y\, e^{y^2}\, dy = \frac{1}{2} e^{y^2} + C$. Pattern: when you see $f'(x)\, e^{f(x)}$, the answer is $e^{f(x)} + C$.

Example 2: u-Substitution with Radicals

Example 2

Evaluate $\int r\sqrt{1+r^2}\,dr$.

This integral appears in surface area calculations (Section 15.5) and cylindrical coordinates (Section 15.7).

Let $u = 1 + r^2$. What is du? Rewrite $\sqrt{1+r^2}$ as $u^{1/2}$.

Choose u: Let $u = 1 + r^2$, so $du = 2r\,dr$, which means $r\,dr = \frac{1}{2}\,du$.

Substitute:

$$\int r\sqrt{1+r^2}\,dr = \int u^{1/2} \cdot \frac{1}{2}\,du = \frac{1}{2} \cdot \frac{u^{3/2}}{3/2} = \frac{1}{3}(1+r^2)^{3/2} + C$$

When you see $r\,dr$ paired with a function of r^2, let $u =$ the function of r^2.

Example 3: u-Substitution with Logarithms

Example 3

Evaluate $\displaystyle\int \frac{\ln y}{y}\,dy$.

This shows up in iterated integrals involving $\ln(y)$ (Section 15.2, homework).

What function has $\frac{1}{y}$ as its derivative? Let $u = \ln y$.

Choose u: Let $u = \ln y$, so $du = \frac{1}{y}\,dy$.

Substitute:

$$\int \frac{\ln y}{y}\,dy = \int u\,du = \frac{u^2}{2} + C = \frac{(\ln y)^2}{2} + C$$

$\displaystyle\int \frac{\ln y}{y}\,dy = \frac{(\ln y)^2}{2} + C$. The "$\frac{1}{y}$" is the derivative of $\ln y$.

Quick Check: u-Substitution

For each integral, identify u, find du, and evaluate.

(a) $\displaystyle\int \cos(3x)\,dx$ (b) $\displaystyle\int x^2 e^{x^3}\,dx$

(c) $\displaystyle\int \frac{e^{\sqrt{t}}}{2\sqrt{t}}\,dt$ (d) $\displaystyle\int_0^1 \frac{2x}{(1+x^2)^2}\,dx$

Show u, du, rewritten integral, and final answer for each.

(a) $u = 3x$: $\frac{1}{3}\sin(3x) + C$

(b) $u = x^3$: $\frac{1}{3}e^{x^3} + C$

(c) $u = \sqrt{t}$: $e^{\sqrt{t}} + C$

(d) $u = 1 + x^2$: $\left[-\frac{1}{u}\right]_1^2 = -\frac{1}{2} + 1 = \frac{1}{2}$

Trig Powers: Using Your Formula Sheet

In polar, cylindrical, and spherical coordinates, you'll constantly encounter powers of $\sin$ and $\cos$. Your formula sheet has the identities — here's how to *use* them.

Even powers ($\sin^2\theta$, $\cos^2\theta$): Use the half-angle identities from your sheet:

$$\sin^2\theta = \frac{1-\cos(2\theta)}{2} \qquad \cos^2\theta = \frac{1+\cos(2\theta)}{2}$$

Odd powers ($\sin^3\theta$, $\cos^3\theta$): Peel off one factor and use $\sin^2 + \cos^2 = 1$:

$$\sin^3\theta = \sin\theta(1-\cos^2\theta)$$

Then $u = \cos\theta$, $du = -\sin\theta\, d\theta$ handles it.

Even powers $\longrightarrow$ half-angle identities. Odd powers $\longrightarrow$ peel off one factor + u-sub.

Example 4: Integrating $\sin^2\theta$

Example 4

Evaluate $\displaystyle\int_0^{2\pi} \sin^2\theta\, d\theta$.

This integral appears in almost every polar and cylindrical coordinate problem (Sections 15.3, 15.7).

Look up the half-angle identity on your formula sheet, substitute, then integrate.

Apply the identity:

$$\int_0^{2\pi} \sin^2\theta\, d\theta = \int_0^{2\pi} \frac{1-\cos(2\theta)}{2}\, d\theta = \frac{1}{2}\left[\theta - \frac{\sin(2\theta)}{2}\right]_0^{2\pi}$$

$$= \frac{1}{2}(2\pi - 0) = \pi$$

$\displaystyle\int_0^{2\pi} \sin^2\theta\, d\theta = \pi$. Similarly, $\displaystyle\int_0^{2\pi} \cos^2\theta\, d\theta = \pi$. These are worth memorizing!

Example 5: Integrating $\sin^3 \phi$

Example 5

Evaluate $\int_0^{\pi/2} \sin^3 \phi \, d\phi$.

This integral appears in spherical coordinate problems (Section 15.8) where the volume element includes $\sin \phi$.

Write $\sin^3 \phi = \sin \phi(1 - \cos^2 \phi)$, then use $u = \cos \phi$.

Peel off one factor:

$$\int_0^{\pi/2} \sin^3 \phi \, d\phi = \int_0^{\pi/2} \sin \phi (1 - \cos^2 \phi) \, d\phi$$

Substitute: $u = \cos \phi$, $du = -\sin \phi \, d\phi$. Limits: $u(0) = 1$, $u(\pi/2) = 0$.

$$= -\int_1^0 (1 - u^2) \, du = \int_0^1 (1 - u^2) \, du = \left[u - \frac{u^3}{3}\right]_0^1 = \frac{2}{3}$$

$\int_0^{\pi/2} \sin^3 \phi \, d\phi = \frac{2}{3}$. “Peel and substitute” works for any odd power.

Integration by Parts

The formula is on your sheet: $\int u \, dv = uv - \int v \, du$

When to use it: The integrand is a **product of two different types** of functions (polynomial $\times$ exponential, polynomial $\times$ trig, etc.).

Choosing u and dv: Use **LIATE** priority (choose u from the first match):

1. **L**ogarithmic: $\ln x$, $\log x$
2. **I**nverse trig: $\arctan x$, $\arcsin x$
3. **A**lgebraic: x^2, x, $3x + 1$
4. **T**rigonometric: $\sin x$, $\cos x$
5. **E**xponential: e^x, 2^x

The function you pick for u gets **differentiated** (simpler). The rest (dv) gets **integrated**. LIATE works for 90% of problems you'll see.

Example 6: Integration by Parts

Example 6

Evaluate $\int x\, e^{2x}\, dx$.

Use LIATE: $u = x$ (algebraic), $dv = e^{2x}\, dx$. Find du and v, then apply the formula.

Set up: $u = x$, $dv = e^{2x}\, dx \Rightarrow du = dx$, $v = \frac{1}{2} e^{2x}$.

Apply the formula:

$$\int x\, e^{2x}\, dx = \frac{x}{2} e^{2x} - \int \frac{1}{2} e^{2x}\, dx = \frac{x}{2} e^{2x} - \frac{1}{4} e^{2x} + C$$

$$= \frac{e^{2x}}{4}(2x - 1) + C$$

$$\int x\, e^{2x}\, dx = \frac{e^{2x}}{4}(2x - 1) + C$$

Example 7: Integration by Parts with $\ln$

Example 7

Evaluate $\int x \ln x \, dx$.

LIATE says $u = \ln x$ (logarithmic beats algebraic). What is dv?

Set up: $u = \ln x$, $dv = x\,dx \Rightarrow du = \frac{1}{x}\,dx$, $v = \frac{x^2}{2}$.

Apply the formula:

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$

$$= \frac{x^2}{4}(2 \ln x - 1) + C$$

When $\ln$ is in the integrand, it's almost always u. Even $\int \ln x \, dx$ uses IBP with $u = \ln x$, $dv = dx$.

Quick Check: Integration by Parts

For each integral, identify u and dv using LIATE, then evaluate.

(a) $\int x \cos x \, dx$ (b) $\int \ln x \, dx$

For (b), try $u = \ln x$ and $dv = dx$. Yes, dv can just be dx!

(a) $u = x$, $dv = \cos x \, dx \Rightarrow v = \sin x$

$$\int x \cos x \, dx = x \sin x - \int \sin x \, dx = x \sin x + \cos x + C$$

(b) $u = \ln x$, $dv = dx \Rightarrow v = x$

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - x + C$$

$\int \ln x \, dx = x \ln x - x + C$ is worth remembering on its own.

Trig Substitution: Start with the Triangle

When you see $\sqrt{a^2 + x^2}$, $\sqrt{a^2 - x^2}$, or $\sqrt{x^2 - a^2}$, **don't memorize substitutions**. Instead, **draw a right triangle** that puts the square root on one side.

The idea: Place a and x on two sides of a right triangle so that the Pythagorean theorem gives you the square root expression on the third side. Then pick θ and read off the substitution.

Example: $\sqrt{4 + x^2}$.

- You need $a^2 + x^2 =$ hypotenuse^2, so put 2 and x on the **legs** and $\sqrt{4 + x^2}$ on the **hypotenuse**.
- Now read: $\tan\theta = x/2$, so $x = 2\tan\theta$, and $\sec\theta = \sqrt{4 + x^2}/2$.

Example: $\sqrt{9 - x^2}$.

- You need $a^2 - x^2$, so put 3 on the **hypotenuse** and x on a **leg**. The other leg is $\sqrt{9 - x^2}$.
- Now read: $\sin\theta = x/3$, so $x = 3\sin\theta$, and $\cos\theta = \sqrt{9 - x^2}/3$.

The triangle **tells you** the substitution — you don't need to memorize anything. Just ask: where do a and x go so the third side is the square root I want?

Example 8: Trig Substitution

Example 8

Evaluate $\displaystyle\int \frac{dx}{\sqrt{4+x^2}}$.

Draw the triangle: legs 2 and x, hypotenuse $\sqrt{4+x^2}$. Read off x in terms of θ, find dx, and simplify.

Draw the triangle: Legs $= 2$ (adjacent) and x (opposite), hypotenuse $= \sqrt{4+x^2}$.
Reading off: $\tan\theta = x/2$, so $x = 2\tan\theta$ and $dx = 2\sec^2\theta\,d\theta$. Also $\sqrt{4+x^2} = 2\sec\theta$.
Substitute:

$$\int \frac{2\sec^2\theta\,d\theta}{2\sec\theta} = \int \sec\theta\,d\theta = \ln|\sec\theta + \tan\theta| + C$$

(The result of $\int \sec\theta\,d\theta$ is on your formula sheet.)

Back-substitute from the same triangle: $\sec\theta = \frac{\sqrt{4+x^2}}{2}$, $\tan\theta = \frac{x}{2}$.

$$= \ln\left|\frac{\sqrt{4+x^2}}{2} + \frac{x}{2}\right| + C = \ln\left|\sqrt{4+x^2} + x\right| + C_1$$

Quick Check: Trig Substitution Setup

For each integral, draw the right triangle first. Then read off the substitution and simplify.

(a) $\displaystyle\int \frac{dx}{\sqrt{9-x^2}}$ (b) $\displaystyle\int \sqrt{1+9x^2}\,dx$

For (a): hypotenuse $= 3$, one leg $= x$, other leg $= \sqrt{9-x^2}$. For (b): rewrite as $\sqrt{1+(3x)^2}$, then draw the triangle.

(a) Triangle: hypotenuse $= 3$, leg $= x$, other leg $= \sqrt{9-x^2}$. Read: $\sin\theta = x/3$, so $x = 3\sin\theta$, $dx = 3\cos\theta\, d\theta$, $\sqrt{9-x^2} = 3\cos\theta$.

$$\int \frac{3\cos\theta\, d\theta}{3\cos\theta} = \int d\theta = \theta + C = \arcsin\left(\frac{x}{3}\right) + C$$

(b) Rewrite: $\sqrt{1+(3x)^2}$. Triangle: legs $= 1$ and $3x$, hypotenuse $= \sqrt{1+9x^2}$. Read: $\tan\theta = 3x$, so $x = \frac{1}{3}\tan\theta$, $dx = \frac{1}{3}\sec^2\theta\, d\theta$, $\sqrt{1+9x^2} = \sec\theta$.

$$\int \sec\theta \cdot \frac{1}{3}\sec^2\theta\, d\theta = \frac{1}{3}\int \sec^3\theta\, d\theta$$

(Use the $\int \sec^3\theta\, d\theta$ formula from your sheet.)

(a) gives $\arcsin$ — that's where the inverse trig formulas *come from*. The triangle approach works every time without memorizing which substitution goes with which form.

Your Turn: Mixed Practice

These are actual integrals from your upcoming homework and exams. Identify the technique and evaluate. Use your formula sheet!

(a) $\displaystyle\int_0^1 2x\, e^{x^2}\, dx$ (b) $\displaystyle\int_0^{\pi} \sin^2\theta\, d\theta$ (c) $\displaystyle\int_0^1 x\, e^{-x}\, dx$

For each: name the technique (u-sub, IBP, or trig identity), then evaluate.

(a) **u-sub** ($u = x^2$): $\left[e^{x^2}\right]_0^1 = e - 1$

(b) **Half-angle** (from sheet): $\int_0^{\pi} \frac{1-\cos(2\theta)}{2}\, d\theta = \frac{\pi}{2}$

(c) **IBP** ($u = x$, $dv = e^{-x}\, dx$): $[-xe^{-x}]_0^1 + \int_0^1 e^{-x}\, dx = -e^{-1} + (1 - e^{-1}) = 1 - \frac{2}{e}$

Step 1 is always: **which technique?** Look for compositions (u-sub), products of different types (IBP), or even powers of trig (identities).

Your Turn: More Practice

Keep going! These all appear in Chapters 15–16.

(a) $\displaystyle\int \frac{x}{\sqrt{1+x^2}}\, dx$ (b) $\displaystyle\int_0^{\pi} \cos^2\theta\, d\theta$ (c) $\displaystyle\int \theta\sin\theta\, d\theta$

Hint: (a) doesn't need trig sub — look for a u-sub first!

(a) **u-sub** ($u = 1 + x^2$, $du = 2x\,dx$):

$$\int \frac{x}{\sqrt{1+x^2}}\,dx = \frac{1}{2}\int u^{-1/2}\,du = \sqrt{1+x^2} + C$$

(b) **Half-angle** (from sheet): $\int_0^{\pi} \frac{1+\cos(2\theta)}{2}\,d\theta = \frac{\pi}{2}$

(c) **IBP** ($u = \theta$, $dv = \sin\theta\,d\theta$):

$$\theta(-\cos\theta) - \int -\cos\theta\,d\theta = -\theta\cos\theta + \sin\theta + C$$

Notice (a): when $x\,dx$ is present alongside $\sqrt{1+x^2}$, u-sub works and trig sub is unnecessary. Always try u-sub first!

When an Integral Can't Be Evaluated

Some functions have **no elementary antiderivative**. No technique on your formula sheet will work:

$$\int e^{x^2}\,dx \qquad \int \cos(y^2)\,dy \qquad \int e^{x^4}\,dx \qquad \int \sqrt{y^3+1}\,dy$$

Why does this matter? In double integrals, you'll sometimes get one of these as an inner integral:

$$\int_0^1 \int_x^1 e^{y^2}\,dy\,dx$$

You **cannot** evaluate $\int e^{y^2}\,dy$ — it's impossible!

The fix: Reverse the order of integration. After switching to $dx\,dy$ and adjusting the limits, the impossible integral disappears and you get something you *can* do (like $\int y\,e^{y^2}\,dy$ — Example 1 from today!).

If an inner integral looks impossible, **don't panic** — reverse the order! We'll learn this technique in Section 15.2.

Which Technique? A Decision Guide

When you see a single-variable integral inside a double or triple integral:

1. Is it a basic antiderivative? (Check your formula sheet.) If yes, just integrate.

2. Is there a composition with its derivative nearby? $\to$ u-substitution.

3. Are there even powers of $\sin$ or $\cos$? $\to$ Half-angle identities (formula sheet).

4. Are there odd powers of $\sin$ or $\cos$? $\to$ Peel one off, then u-sub.

5. Is it a product of different function types? $\to$ Integration by parts (LIATE).

6. Is there $\sqrt{a^2 \pm x^2}$ or $\sqrt{x^2 - a^2}$ without a helpful $x\,dx$? $\to$ Trig substitution (table on sheet).

7. None of the above work? $\to$ The integral may be impossible. Reverse the order of integration!

Always try u-sub first. It's the right move about 70% of the time.

Homework

Section 15.1b: Practice the integrals on this handout!